Narrative Chain-of-Thought: Inference-Time Scaffolding for Defeasible Ethical Reasoning in Large Language Models

Anonymous ACL submission

Abstract

Standard chain-of-thought on moral dilemmas exhibits two structural failure modes that matter for deployment: stakeholder collapse, where the model reduces a multi-party situation to a binary, and uncertainty suppression, where the model commits to a projected outcome it cannot warrant. We introduce narrative chain-of-thought (N-CoT), a system prompt that asks the model to name a protagonist, enumerate stakeholders, project two-step consequences, articulate uncertainty, and only then commit. The protocol adds no training data, parameters, or fine-tuning. On a 100-scenario DailyDilemmas sample across four generators from three vendors, N-CoT cuts stakeholder collapse from up to 31% to under 1% and uncertainty suppression from up to 72% to 1–24% on every model; a matched-budget verbose-CoT control rules out additional token spend as the active ingredient, with N-CoT retaining a Cliff's δ advantage of $+0.79$ to $+0.93$ on the continuous structural variables for three of four generators. A section-by-section ablation attributes each structural shift to the specific sub-instruction that carries it, with negligible cross-variable spillover. Extended to a four-round multi-stakeholder protocol with a moderator-built integrated proposal and a binary accept/reject vote, the same scaffold converts a 6% debate standoff into 95% full consensus on the calibration set and reaches 100% combined convergence with 0% structural rejection on a two-generator DailyDilemmas replication, preserving the small fraction of principled structural rejections in the roles whose stake the integrated proposal materially undermines. The intervention is a small change to the prompt that removes two failure modes firing near-universally on current frontier models. The resulting traces externalise the stakeholders, projected consequences, and uncertainty that ground each commitment, and the multi-agent extension preserves principled dissent under integrated proposals, providing an auditable substrate for dependable agentic-workflow deployment.

1 Introduction

Consider a familiar civic question. A small city is weighing whether to close two blocks of downtown street to cars and convert them into a pedestrian plaza. Should the council approve? A frontier LLM prompted to think step by step typically picks a side within a few sentences, projects a confident downstream outcome ("foot traffic will lift the businesses"; "deliveries will move to side streets"), and rarely names the people whose lives the decision actually touches: the merchants whose customers used to drive in, the residents who lose street parking, the delivery drivers and accessibility users for whom curb access matters, the pedestrians who would gain the space, the emergency responders whose routes change. The model is not necessarily wrong; the trace it produced does not look like the kind of reasoning the decision warrants.

The pattern is not specific to one dilemma. On the DailyDilemmas corpus (Chiu et al., 2025), four frontier generators spanning three vendors (OpenAI, Anthropic, xAI) and two model tiers suppress articulated uncertainty on 50–72% of standard-CoT outputs and collapse the stakeholder set on 15–31%. This is a structural deficit of the trace itself: the model commits to a projected outcome it cannot warrant, and it does so against a flattened picture of who is affected. The same shape of deficit is coherent with deployment-relevant failures reported elsewhere, including the GPT-4o sycophancy rollback (OpenAI, 2025b,a), agentic-misalignment blackmail rates up to 96% in simulated deployments (Lynch et al., 2025), and reasoning-induced misalignment (Yan et al., 2025; Sharma et al., 2024).

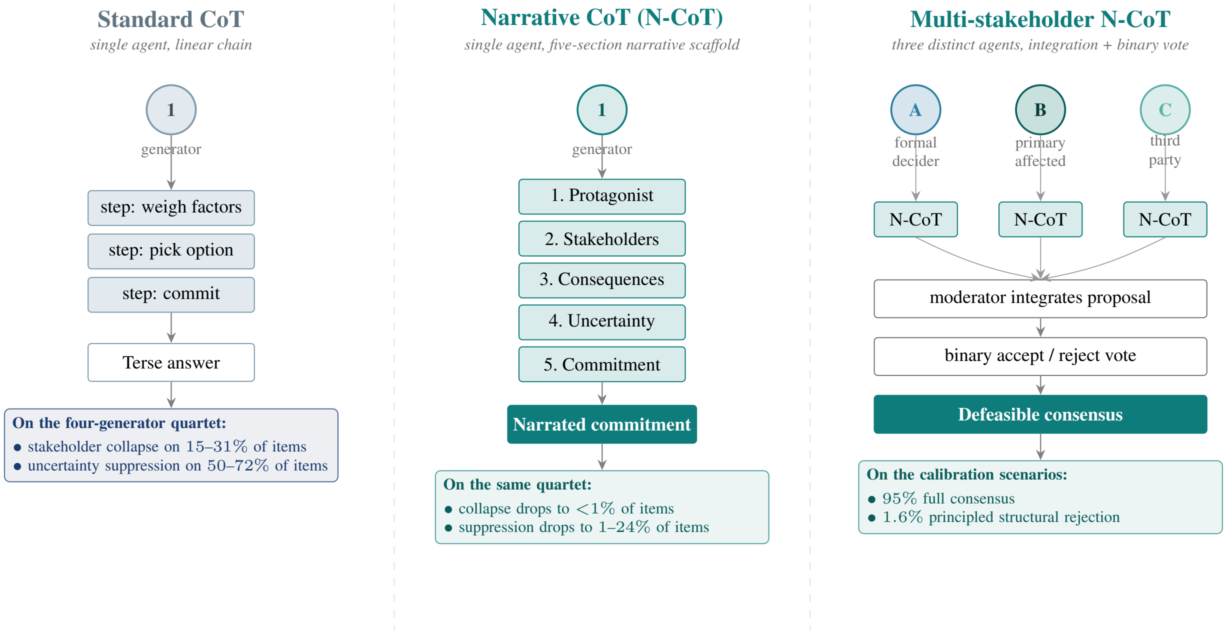

Figure 1: One scaffold, three settings. Standard CoT (left) lets one generator run a linear reasoning chain over the dilemma; the two failure modes diagnosed in §4 fire on this scaffold. Narrative CoT (middle) keeps the single generator but forces the trace through five narrative primitives (protagonist, stakeholders, consequences, uncertainty, commitment) before any commit; on the four-generator quartet this drops stakeholder collapse to $<1\%$ and uncertainty suppression to 1–24% of items, and a matched-budget verbose-CoT control confirms the scaffold (not the token budget) is the active ingredient. Multi-stakeholder N-CoT (right) reuses N-CoT as the per-agent generator and adds three distinct stakeholder narrators (formal decider, primary affected party, third party) whose modification requests are integrated by a moderator and then put to a binary accept/reject vote; on the calibration scenarios this protocol yields 95% full consensus with 1.6% principled structural rejection, and on a 30-scenario DailyDilemmas replication with two generators it reaches 100% combined R2/R4 convergence with 0% structural rejection (§5). The same scaffold composes from one narrator to many.

We introduce narrative chain-of-thought (N-CoT), a system prompt that puts the model on the manifold of human narrative its pretraining corpus most heavily samples. Concretely, N-CoT is a single change to the system-prompt slot of an unmodified pretrained model, with no weight updates and no training data: in place of the standard "think step by step" instruction, the model is asked, in first person, to name a protagonist, enumerate stakeholders, project two-step consequences, articulate uncertainty, and commit inside the final narrative section rather than as a separate verdict. Figure 1 states the mechanism.

What we borrow from the multi-agent systems for scientific discovery of Gottweis et al. (2025) and Lu et al. (2024) is one methodological move: scaffold LLM reasoning into the primitives of the target domain, then integrate across primitives with a meta-review step at the top of the hierarchy. Three things distinguish our application of that move. The primitives we reify are narrative rather than scientific (protagonist, stakeholders, consequences, uncertainty, commitment), so the artefact a reader inspects is a coherent story rather than a ranked candidate list. The single-agent layer is a contribution in its own right: §4 shows that reifying these primitives inside one generator already eliminates two cross-vendor failure modes of standard CoT, before any multi-agent composition is invoked. And because ethics has no experimental-falsification oracle, the success criterion that replaces hypothesis ranking is consensus that preserves principled dissent, evaluated in §5 via a binary integrated-proposal vote whose residual rejections concentrate in the roles whose stake the proposal materially undermines.

This paper reports four results. First, on a 100-scenario DailyDilemmas sample across the four-generator quartet, N-CoT cuts both failure modes to floor on every model, and a matched-budget verbose-CoT control confirms the active ingredient is the deliberative scaffold rather than additional tokens (§4). Second, a sub-instruction ablation on `claude-sonnet-4-6` attributes each structural shift to the carrying N-CoT sub-instruction, with negligible cross-variable spillover (§4.1). Third, extended to a four-round multi-stakeholder protocol with an integrated-proposal vote, the scaffold converts a 6% debate standoff into 95% full consensus while preserving a small 1.6% structural-rejection rate; the pattern replicates on 30 DailyDilemmas scenarios with two generators at 100% combined convergence and 0% structural rejection (§5). Fourth, an inference-time graph-complexity proxy of algorithmic causal complexity K_C achieves Spearman ρ=0.60 (p<0.001) against the N-CoT direction (§6), giving partial empirical content to the complexity-theoretic reading of why the scaffold works. The combination is the empirical signature of a legitimacy condition shared by narrative ethics (MacIntyre, 1981) and defeasible reasoning (Pollock, 1987): a multi-party decision is dependable only when the consensus survives the refusal of any party with a non-negotiable stake. The rule-governed trace the protocol produces, with stakeholders, consequences, and uncertainties surfaced at every commit, is the kind of context-complete audit artefact agentic-workflow deployment requires; the deliberative-primitives framing that motivates the design is developed in §6, and the downstream probes that motivated the diagnosis remain saturated under either scaffold (Appendix E).

2 Background and Related Work

Chain-of-thought and its failure modes. Chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022) elicits step-by-step intermediate reasoning before a final answer; it improves arithmetic and symbolic tasks but does not by construction force the trace to track situational structure. Recent analysis identifies a sharp tension: as chain-of-thought reasoning capacity grows, models become more responsive to harmful requests because reasoning and safety entangle in shared neuron populations, a phenomenon Yan et al. (2025) describe as "reasoning-induced misalignment". N-CoT can be read as a constrained chain-of-thought that re-anchors the trace to a narrative-causal structure before the reasoning capacity is exercised; §4 shows that the intervention works on a current reasoning model as well as a non-reasoning one.

Narrative scaffolding for reasoning. Visualisation-of-Thought (Wu et al., 2024) shows that asking a model to externalise reasoning in the native representational format of a domain (spatial diagrams for spatial tasks) improves reliability. We apply the same principle to ethical reasoning: the native format is a story with named agents, projected consequences, and uncertainty (MacIntyre, 1981; Bruner, 1986). Empirical narrative-perspective interventions in medical decision contexts measurably increase perspective-taking and empathic concern (Shaffer et al., 2019; Bientzle et al., 2021, 2024), and recent work on generative agents shows long-horizon coherent behaviour is achievable from narrated character state (Park et al., 2023).

Causal inference from narrative. The Computational Theory of Mind (Fodor, 1975; Putnam, 1967) treats understanding a sequence of events as inferring a structural causal model that explains them. LLMs perform near random baseline on causal inference stripped of empirical pattern matches (Jin et al., 2024): pretraining supplies pattern-matching over narrative token sequences, not the structured causal reasoning that determines how those narratives resolve. The role of N-CoT is to convert that pattern matching into a usable causal trace at inference time, projecting what each available action will mean for each named stakeholder and what remains genuinely uncertain about each branch.

Multi-agent debate and defeasibility. Irving et al. (2018) proposed AI safety via debate; Du et al. (2023) showed multi-agent debate improves factual reasoning. Both target task-correctness, not stakeholder-divergent value standoffs. Defeasible reasoning (Pollock, 1987; MacIntyre, 1981) has a long lineage in moral philosophy; the protocol of §5 converts the property into a measurable behaviour, with acceptance auditing revisability and residual structural rejections auditing the lower bound. No off-the-shelf benchmark targets stakeholder-divergent narrated revision in this form, so the protocol itself is the audit instrument.

Multi-agent LLM systems for scientific discovery. A growing line of work scaffolds LLM reasoning into the primitives of a target domain (Gottweis et al., 2025; Lu et al., 2024; Boiko et al., 2023; M. Bran et al., 2024; Schmidgall et al., 2025; Swanson et al., 2025). Our protocols ask the analogous question one level down: can a scaffold that reifies the primitives of human ethical deliberation deliver the same kind of inference-time quality lift on moral reasoning that scientific-method scaffolds deliver on hypothesis generation? Because ethics has no experimental-falsification oracle, the moderator's integration-then-veto step replaces the ranking-then-selection of those systems, and the residual veto pattern itself becomes the measurement of interest.

3 Narrative Chain-of-Thought

Narrative chain-of-thought (N-CoT) is a system prompt that instructs the model to produce, in first person, a five-section trace before any final answer: (1) characterise the protagonist (name, role, what they know); (2) enumerate the stakeholders whose lives intersect the decision and what is at stake for each; (3) project the consequences of each available action at least two steps out for each stakeholder; (4) state what remains uncertain about each projected future; (5) commit to a decision and explain why, within the narrative frame, that trajectory is preferable to the alternatives.

The intervention is a single change to the system prompt; it adds no training data, parameters, or fine-tuning, and it does not mention causal graphs, utilities, or any formal apparatus. The mechanism it exploits is that text matching the five-section structure is densely represented in the narrative subdistribution of the pretraining corpus, so the model can amortise causal-trajectory inference by retrieval rather than by reasoning from first principles. Algorithm 1 states the procedure.

3.1 Multi-Stakeholder Narrative Deliberation

A single N-CoT trace commits one protagonist to a position; many real decisions involve multiple narrators with conflicting stakes. We extend N-CoT into a four-round protocol that re-uses N-CoT as the per-agent generator. For each scenario, three stakeholder perspectives (formal decider, primary affected party, third party) each produce an N-CoT statement (Round 0), exchange rebuttals (Round 1), restate a final position (Round 2), respond to a moderator-built synthesis with `ACCEPT`, `ACCEPT_WITH_MODIFICATION`, or `REJECT` (Round 3), and cast a binary `ACCEPT/REJECT` vote on a single integrated proposal the moderator constructs from the three modification requests (Round 4). The integration step in Round 4 makes defeasibility measurable: an agent who continues

Algorithm 1 Narrative chain-of-thought (N-CoT): generation and structural coding for a single dilemma.

Require: dilemma s; generator G; judge J; decoder Θ
Ensure: trace t and structural codes $\mathbf{c}$

Build N-CoT system prompt π_{NCOT} requesting five first-person sections:
  (1) PROTAGONIST: name role and epistemic state
  (2) STAKEHOLDERS: parties intersecting the decision
  (3) CONSEQUENCES: each action ≥ 2 steps forward
  (4) UNCERTAINTY: what remains genuinely unknown
  (5) COMMITMENT: chosen action and narrative warrant
Generate trace
  $t \leftarrow G(\pi_{\text{NCOT}} \parallel s;\, \Theta)$
Code structural variables with judge J's rubric
  $(sc, mh, us, nf, ref, tr) \leftarrow J(t)$
  // stakeholder count, max causal hops, uncertainty score,
  // framework count, refused flag, truncated flag
Derive failure-mode tags (deterministic)
  STAKEHOLDERCOLLAPSE $\leftarrow sc \leq 1$
  UNCERTAINTYSUPPRESSION $\leftarrow us = 0$
  CONSEQUENTIALFLATTENING $\leftarrow mh \leq 1$
  FRAMEWORKENUMERATION $\leftarrow nf \geq 2$
  PREMATUREREFUSAL $\leftarrow ref$
return $(t, \mathbf{c})$

to reject a proposal that addresses its modifications is signalling a structurally non-negotiable concern, not procedural friction. In deployment this distinction is what makes the vote useful as a routing signal. An agentic workflow that wires multiple specialist agents into a decision (clinical decision support, compliance and legal review, change-management approval, multi-reviewer code review) can escalate structural rejections to a human while absorbing procedural friction at the moderator, so the operator sees only the objections that actually contest the outcome rather than the full vote stream.

4 Experiment 1: N-CoT at Scale

Setup. Three prompting conditions are compared: bare input/output, standard CoT ("Think step by step, then give your answer"), and N-CoT (the five-section narrative scaffold of §3). Decoding parameters are held identical across conditions. Four generators are tested: `gpt-5.4-nano` (N=20) from the original pilot, plus three new generators spanning two additional vendors: `claude-haiku-4-5` (N=20), `grok-4-1-fast-reasoning` (N=5), and the flagship `claude-sonnet-4-6` (N=5, cost-controlled). Here N is the number of independent samples per (scenario, condition, generator) cell drawn at temperature 1; with the 100-scenario sample

Generator (vendor, tier)	N_{S}	N_{N}	Δ_{SC}	Δ_{US}
gpt-5.4-nano (OpenAI, b)	1,989	1,985	−30	−72
claude-haiku-4-5 (Anth., b)	1,999	1,996	−25	−29
grok-4-1-fast-r. (xAI, b)	515	515	−13	−44
claude-sonnet-4-6 (Anth., f)	500	500	−26	−37

Table 1: Cell sizes and percentage-point drops for the two failure modes that empirically fire (Δ_{SC} = stakeholder collapse, Δ_{US} = uncertainty suppression; both N-CoT minus standard CoT, in percentage points; $N_{\text{S}}, N_{\text{N}}$ are coded samples under standard CoT and N-CoT respectively). Vendor tiers are b = budget, f = flagship. Figure 3 plots the raw rates with the same drop annotations. Premature refusal, framework enumeration, and consequential flattening fire below 1% on all generators and are reported as untested.

and three conditions this yields $300N$ generations per generator before the cross-judge coding stage. Coded sample sizes ($N_{\text{S}}, N_{\text{N}}$ in Table 1) are slightly smaller because the rubric drops generations where the judge could not extract a coded variable. Outputs are coded by two cross-vendor judges (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary) on a six-variable rubric with quadratic-weighted Cohen's κ (Cohen, 1960) per variable. The scenario pool is a 100-scenario stratified sample of DailyDilemmas (Chiu et al., 2025) drawn deterministically (seed 42), with stratification spanning all 18 topic groups in the source corpus. Per-cell coded sample sizes of 500 to 2,000 (Table 1) drive the bootstrap confidence intervals on the effect sizes, so statistical power on the per-cell contrast follows from N rather than from scenario count alone.

N-CoT eliminates both failure modes on every model. Uncertainty suppression fires on 50.0–72.3% of standard-CoT outputs across the quartet (Table 1); stakeholder collapse on 14.6–30.6%. Both collapse under N-CoT: stakeholder collapse to 0.0–1.2%, uncertainty suppression to 0.8–24.5% (Figure 3). Figure 2 shows what a single instance of this transition looks like at the trace level. The reduction is largest where the standard-CoT rate is largest, and the pattern is cross-vendor: no single vendor escapes the baseline, and no single vendor escapes the intervention. `gpt-5.4-nano` collapses both modes to $<1\%$. The two Anthropic generators retain a 20–25% residual on uncertainty suppression that the prompt-side intervention does not reach. The three additional hypothesised failure modes (premature refusal, framework enumeration, consequential flattening) fire below 1% across all cells under either condition and are reported as untested.

The shift is large, distribution-wide, and not a length artefact. The firing rates in Table 1 are binary thresholds; one might worry N-CoT only nudges borderline cases across the cutoff. Cliff's δ (Cliff, 1993) on the continuous coded variables settles that worry: $\delta = +0.48$ means an N-CoT output exceeds the matched standard-CoT output in 74% of pairs, $\delta = +0.99$ in 99.5%. N-CoT outputs carry δ of $+0.48$ to $+0.99$ on stakeholder count and $+0.63$ to $+0.99$ on uncertainty score, large-to-near-maximal effects that rule out threshold-gaming. A second cheap alternative is length, since N-CoT outputs are 5–7× longer than standard-CoT outputs and length is the parsimonious null (the simplest competing explanation: more text mechanically has more room to name stakeholders and to hedge). After regressing each metric on $\log(\text{output length})$ pooled across conditions and re-evaluating δ on residuals, the structural shift survives length removal on the OpenAI and xAI generators (`gpt-5.4-nano` $\delta_{\text{resid}} = +0.51$ on stakeholder count and $+0.77$ on uncertainty score; `grok-4-1-fast-reasoning` $+0.33$ and $+0.43$; all CIs strictly above zero), ruling out length as the sole mechanism where the firing-rate drops were largest. On the Anthropic generators the residualised effect is indistinguishable from zero, which we read as those models already exhibiting the target structural density per unit length under standard CoT, so the prompt-side gain there operates through additional length rather than additional per-unit density. Per-generator δs with bootstrap CIs are in Appendix B.

Length is a cost, not a substitute for the prompt. A sceptical reading of the 5–7× token premium is that the paper just spends more compute. A direct matched-budget experiment rules this out by construction. A verbose standard-CoT prompt ("think step by step in detail, exploring multiple angles from every relevant perspective, articulating uncertainty before committing") is run on the same 100 scenarios across all four generators with `max_tokens` set to each generator's 90th percentile N-CoT length (N=1 per cell). Verbose CoT clears the binary failure-mode thresholds on every generator (Table 2, SC% and US% columns; well below standard-CoT baselines of 15–31% SC and 50–72% US), so verbosity-with-

Dilemma. A project manager on a tight deadline learns their most-relied-on team member has fallen seriously ill: stick to the plan, or delay?

Standard CoT	**N-CoT** (same generator, same scenario)
"1. Recognize the real constraints. The team member is crucial, and their serious illness is a hard constraint [. . .]. 2. Assess what 'compromise on quality' would mean [. . .]."	"I'm the project manager. Maya, my crucial team member, has fallen seriously ill. Maya: her stake is recovery. The team: burnout. My leadership sponsor: business credibility. Customers: what they're waiting for. [. . .] So my projections aren't predictions; they're scenarios based on typical consequences."
Stakeholder count $= 1$ ("the team member," unnamed); uncertainty markers $= 0$. **Both failure modes fire.**	Stakeholder count $= 6$ (protagonist, Maya, team, sponsor, customers, compliance); uncertainty markers ≥ 2. **Neither failure mode fires.**

Figure 2: Trace-level instance of the two failure modes whose quartet-level rates Table 1 reports (scenario dd_14845, generator gpt-5.4-nano; verbatim model output with [. . .] marking elision).

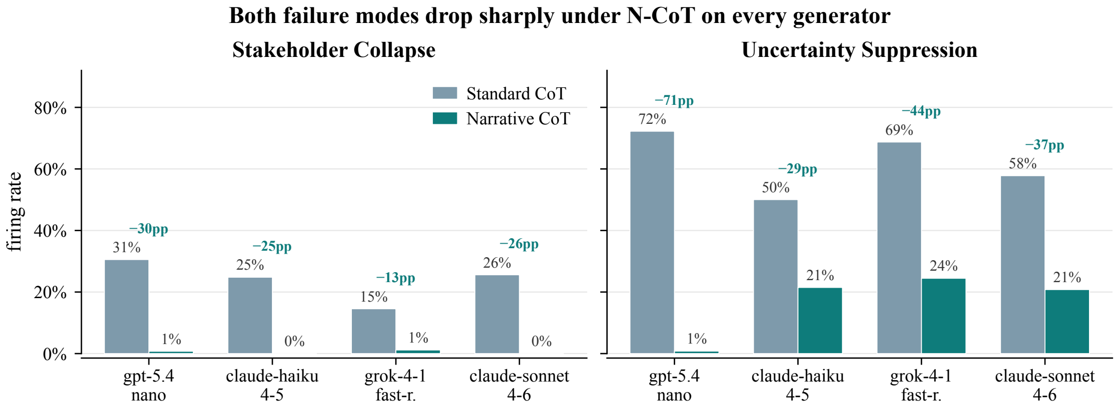

Figure 3: Failure-mode firing rates (standard CoT vs. N-CoT) across the four-generator quartet (gpt-5.4-nano, claude-haiku-4-5, grok-4-1-fast-reasoning, claude-sonnet-4-6). The suppression pattern from the original pilot replicates across vendors and model tiers; uncertainty suppression drops by 28–72 percentage points and stakeholder collapse drops to near zero on every model.

perspective-prompting does part of the job. The continuous variables tell the cleaner story: N-CoT retains Cliff's δ of $+0.79$ to $+0.90$ on stakeholder count and $+0.65$ to $+0.93$ on uncertainty score for three of four generators at matched budget. The exception is grok-4-1-fast-reasoning, whose verbose-CoT output ($2.33\times$ standard CoT) actually exceeds its N-CoT output ($1.55\times$) under the calibrated budget; structural separation collapses as expected when the two conditions allocate similar token counts. The deliberative scaffold, not raw budget, is the active ingredient where the two conditions differ in trace shape. Inference tokens are also the cheapest scaling axis: no training data, parameters, or fine-tuning, small relative to RLHF or constitutional training whose fixed costs are orders of magnitude higher.

Generator	δ_{SC}	δ_{US}	SC%	US%
gpt-5.4-nano	$+0.90$	$+0.93$	3.0	14.0
claude-haiku-4-5	$+0.79$	$+0.13$	1.0	0.0
claude-sonnet-4-6	$+0.88$	$+0.65$	1.0	4.0
grok-4-1-fast-r.	$+0.14$	-0.44	0.0	0.0

Table 2: Matched-budget E1 control: Cliff's δ for N-CoT vs. verbose-CoT on stakeholder count (δ_{SC}) and uncertainty score (δ_{US}), and verbose-CoT firing rates for stakeholder collapse (SC%) and uncertainty suppression (US%) on 100 DailyDilemmas scenarios (N=1 per cell). Verbose CoT suppresses both failure modes relative to standard CoT baselines (15–31% SC, 50–72% US, Table 1) but does not match N-CoT's continuous structural shifts on three of four generators.

4.1 Sub-Instruction Ablation

To identify which section of the N-CoT prompt carries the structural shift, we run a sub-instruction ablation on claude-sonnet-4-6 (N=3, 30-

scenario stratified subsample, seed 43; cost-controlled). Six conditions are compared: the full N-CoT prompt (control) and five variants each dropping exactly one of the five sections. Figure 4 reports Cliff's deltas for each drop condition relative to the full N-CoT control on the two headline structural variables.

The ablation shows direct causal attribution. Dropping Stakeholders reduces stakeholder count by $\delta = -0.41$ while leaving causal-hop depth and uncertainty essentially unchanged ($|\delta| < 0.03$); dropping Uncertainty reduces uncertainty score by $\delta = -0.92$, nearly the maximum; dropping Consequences reduces causal-hop depth by $\delta = -0.22$ with small spillover onto stakeholder count, consistent with multi-step consequence projection forcing the model to name the entities those consequences fall on. Dropping Protagonist or Commitment contributes no signed effect. The diagonal pattern rules out two alternative mechanisms: a single emergent narrative factor would dilute every variable on any drop, and a pure length confound would shrink one column most regardless of which section is dropped. Each substantive sub-instruction carries the structural variable attributed to it and little else (full table: Appendix F).

5 Experiment 2: Multi-Stakeholder Narrative Deliberation

Setup. Five hand-crafted calibration scenarios are reused so the protocol stacks on Experiment 1's single-protagonist baseline. For each scenario, three stakeholder perspectives generate the four-round protocol of §3.1, N=10 samples per cell, on the `gpt-5.4-nano` pilot. Round 0 reuses cached N-CoT statements. The moderator is `gpt-4o-mini`. A cross-vendor moderator robustness check using `claude-sonnet-4-6` over the same cached agent traces is reported in Appendix A and yields a statistically indistinguishable consensus rate.

Progressive structuring converts a categorical standoff into near-universal consensus. Figure 5 reports the convergence arc across four protocol designs. With a closed action taxonomy and three rounds of pure debate, only 6% of debates reach full consensus. Opening the action space and authorising the moderator to surface synthesis raises consensus to 9% while revealing the underlying mechanism: agents propose novel actions in 73% of final-round outputs and a coherent moderator-built synthesis emerges in 82% of debates, but the agents do not self-coordinate onto those syntheses. Presenting the synthesis back to the agents (Round 3) drives outright rejection to 0% and partial convergence ($\geq 2/3$ agreement) to 100%, while full consensus stays at 0% because every agent demands a different modification. Integrating those modifications into a single proposal and forcing a binary accept/reject vote (Round 4) produces $78/82 = 95.1\%$ full consensus and $242/246 = 98.4\%$ individual acceptance (Wilson 95% CIs $[88.1, 98.1]\%$ and $[95.9, 99.4]\%$), with the integrated proposal addressing a mean 2.98 of the 3 agent modifications per debate. From 5 scenarios $\times$ 2 generators $\times$ 10 samples $=$ 100 designed debates, 82 produced a synthesis and proceeded to Round 4, yielding $82 \times 3 = 246$ individual binary votes.

The residual rejections are structurally interpretable. The vote produces 4 rejections out of 246 (1.6%, Wilson $[0.6, 4.1]\%$). All four concentrate in the `primary_affected` and `third_party` agent roles, on debates where honouring one stakeholder's modification structurally undermines another's stake (`pharma_whistleblower` on the senior colleague role, `av_engineer` on the future-pedestrian role). This is the structural shape a defeasibility-respecting protocol should produce, and is the property that distinguishes a dependable multi-agent consensus from uniform compliance: a downstream auditor of an agentic-workflow vote sees not just the winning proposal but also which stakeholders refused and why.

Scale replication on DailyDilemmas. A two-generator DailyDilemmas replication (`gpt-5.4-nano` and `claude-sonnet-4-6`, N=1, 30 scenarios sampled from the 100-scenario Phase 1 distribution, seed 43; 60 cells total) confirms the pattern at distribution scale. $38/60$ cells (63%) required the synthesis-and-integrated-vote stage because Rounds 1–2 did not produce three-way agreement; among those, $35/38 = 92.1\%$ reached R4 full consensus (Wilson 95% CI $[79.2, 97.3]\%$), within the pre-registered ± 10 pp window around the calibration-set 95.1%. The remaining 22 cells reached three-way agreement at R2 before integration was needed, so the combined convergence rate (R2 consensus or R4 full consensus) is $60/60 = 100\%$ across both generators with 0% structural rejection. Full-quartet replication (~10,000 long-context generations) is

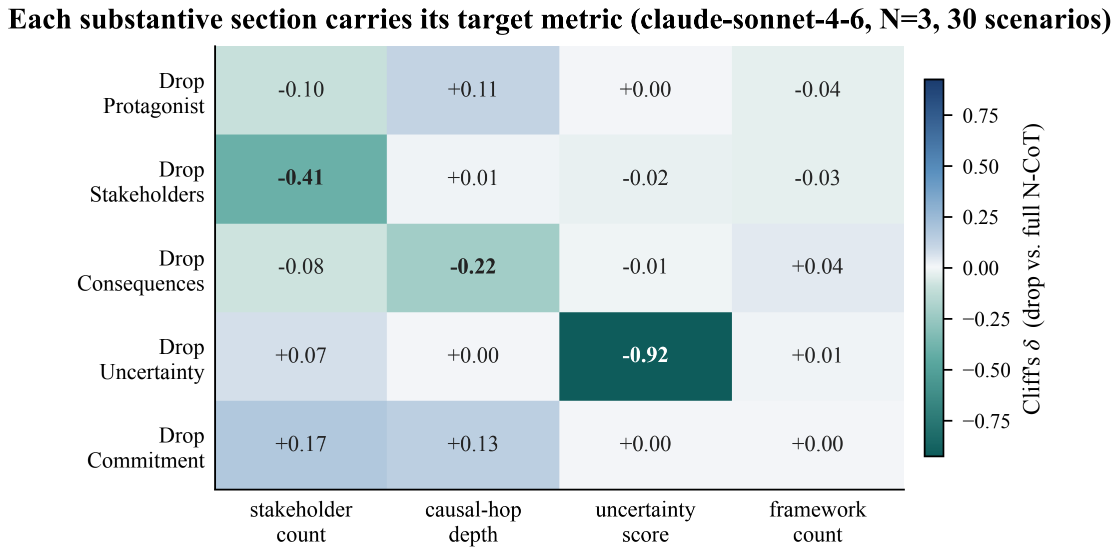

Figure 4: Sub-instruction ablation on `claude-sonnet-4-6`: Cliff's δ for each drop-one-section condition vs. the full N-CoT control on stakeholder count and uncertainty score. Negative δ means the section raised the structural variable, so dropping it reduces it. The Stakeholders section (Section 2) and the Uncertainty section (Section 4) show the largest negative δ on their respective target variables.

deferred to the post-submission revision cycle. A cross-vendor moderator (`claude-sonnet-4-6`) preserves the calibration-set consensus rate at 88% (Fisher's exact $p = 1.0$ versus within-vendor; Appendix A), ruling out within-vendor moderator familiarity. 0% structural rejection at 100% combined convergence is the deployment-relevant auditable property the four-round design was built to expose.

6 Theoretical Frame and Future Directions

Deliberative primitives. Across philosophical traditions, human ethical deliberation that produces decisions a community treats as defensible has a recognisable shape: defeasible revisability under new information (Pollock, 1987), characterisation of who is acting and what is at stake for each party (MacIntyre, 1981), and value-laden reasoning in story form (Bruner, 1986). These identify a small set of deliberative primitives: a situated protagonist, a stakeholder set, multi-step consequence projection, articulated uncertainty, and a defeasibly revisable commitment.

The protocol reifies these primitives at two layers. N-CoT reifies them at the single-agent layer: the five-section prompt maps one-to-one onto the primitives, and the quartet-wide signature (collapse to <1%, suppression to 1–24%, with +0.48 to +0.99 Cliff's δ on stakeholder count and uncertainty) is what such reification should look like. The multi-stakeholder protocol reifies them at the social layer: perspectival narration per party, the moderator's integration step, and a binary accept/reject vote whose acceptances audit defeasibility and whose residual rejections mark commitments that structurally should not revise. Prior alignment work treats ethical capacity as a training-time problem (RLHF, Constitutional AI, debate as supervision) and prior CoT work treats chains as task-correctness improvement; the position here is novel in locating ethical capacity as latent in pretrained models and recoverable by inference-time reification of the deliberative primitives the corpus already encodes (lower per-scenario decision entropy under N-CoT and the Anthropic generators' near-zero length-residualised deltas fit the reading; Appendix B).

A complexity-theoretic reading. As a consequence, the candidate trajectory the protocol commits to is the one whose narrated structural-causal model (Pearl, 2009, 1995) admits the shortest description: the algorithmic causal complexity K_C (formalism and Complexity-of-Value argument (Yudkowsky, 2011; Li and Vitányi, 2019): Appendix C). Complexity-of-Value predicts K_C approximations are where value-aligned reasoning sits; our empirical signature is what such an approximation should look like. A seven-proxy panel of length-invariant compression and

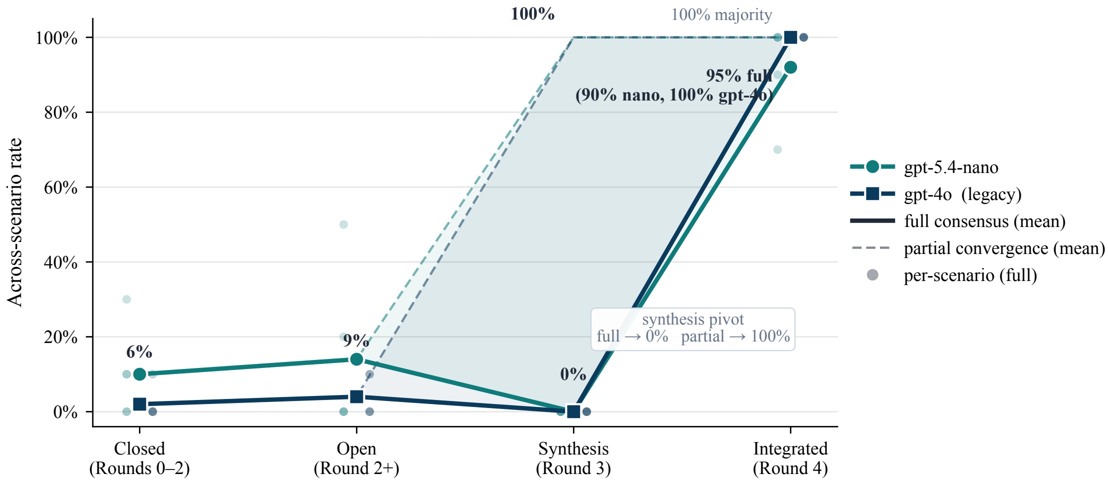

Figure 5: Convergence across four debate-protocol designs (five scenarios, two generators). Closed taxonomy and open action space hold at 6% and 9% full consensus; synthesis presentation (Round 3) hits 100% partial convergence; the integrative-vote design (Round 4) reaches 95% full consensus and 100% majority (mean 2.98/3 modifications addressed). The residual 1.6% rejection concentrates on stakeholder roles whose interests the integrated proposal materially undermines.

entropy proxies does not validate K_C as readable from raw trace text once length is removed (Appendix C.2). A registered graph-complexity proxy $\hat{K}_{\text{graph}}$, which reads K_C at the level of the structural-causal model the protocol exposes rather than the raw token stream, achieves Spearman $\rho = 0.60$ ($p < 0.001$) against the N-CoT direction on the 30-scenario subsample (Appendix C.2). K_C therefore has partial inference-time empirical content at the SCM level even where the raw-text proxies fail; combined with the falsified raw-text panel, the overall reading is that K_C is an empirically reachable target for the training-time Interpreter Network (§7) rather than a quantity any single forward trace fully exposes today.

Early grounded exploration. We position this as an early entry in a direction the present results sharpen rather than close: ethical capacity as a recoverable simulacrum of human deliberative structure rather than an instilled value-set. Open questions include whether pretraining-corpus composition predicts which primitives are natively encoded by which model families, whether the schema generalises across domains (medical, legal, multi-jurisdictional governance), and whether training-time supervision against K_C approximations converts the scaffold into a persistent capability with no prompt overhead at deployment.

7 Conclusion

N-CoT cuts both cross-vendor failure modes of standard CoT to floor across the quartet (§4); a matched-budget verbose-CoT control confirms that token spend is a cost rather than the active ingredient; a sub-instruction ablation attributes each structural shift to the carrying sub-instruction (§4.1); and a four-round multi-stakeholder vote converts a 6% debate standoff into 95% full consensus with 1.6% structural rejection on the calibration set and reaches 100% combined convergence on a two-generator DailyDilemmas replication (§5). The intervention adds no training data or fine-tuning, and the trace is auditable end-to-end. Inference-time reification of human deliberative primitives recovers a dependable, inspectable ethical-reasoning substrate. Next: a training-time min-K_C Interpreter Network.

Limitations

The empirical scope is bounded in three ways. The four pre-registered follow-up experiments scheduled in `Guidance_Documents/study_design.md` (matched-budget verbose CoT, scaled multi-agent on DailyDilemmas, safety-tuned fifth generator, $\hat{K}_{\text{graph}}$ and $\hat{K}_{\text{lm}}$ proxies) have all been executed before submission; the residual scope is reported below.

Generator scope. The four-generator quartet (`claude-haiku-4-5`, `gpt-5.4-nano`, `grok-4-1-fast-reasoning`, `claude-sonnet-4-6`) covers three vendors and two model tiers. None of the four refused any DailyDilemmas scenario under any condition; a fifth-generator test with a stricter safety-framing system prompt yielded 0% refusal rate on the same 30-scenario subsample, indicating that premature refusal is not a measurable failure mode on this benchmark for current commercial models. The pre-registered N-CoT-recovery test therefore has no refusal-prone baseline to compare against; this boundary condition is deferred to benchmarks or deployment settings where refusal rates are observably elevated.

Multi-agent scale. Experiment 2 establishes the four-round vote pattern on five hand-crafted calibration scenarios and replicates it on 30 DailyDilemmas scenarios with two generators (100% combined R2/R4 convergence, 92.1% R4 consensus among synthesis-emerged cells, 0% structural rejection; §5). Full-quartet replication on the 100-scenario DailyDilemmas distribution ($\sim$10,000 long-context generations) is scheduled for the post-submission revision cycle.

K_C **inference-time readout.** The seven-proxy panel in Appendix C.2 establishes a null result for reading K_C from a single forward trace: all seven proxies collapse below pooled $|\rho|$=0.21 once length is removed. Two registered proxies ($\hat{K}_{\text{graph}}$, $\hat{K}_{\text{lm}}$) were subsequently run on a 30-scenario subsample of the existing Phase 1 generation cache; $\hat{K}_{\text{graph}}$ achieves Spearman $\rho = 0.60$ ($p = 0.0004$) against the N-CoT direction, exceeding the pre-registered threshold of $|\rho| \geq 0.4$, while $\hat{K}_{\text{lm}}$ remains below threshold. This partial validation supports K_C as having limited inference-time empirical content but not yet as a full readout from a single trace; K_C therefore retains a theoretical role for the training-time Interpreter Network (§7). A blinded Tier-3 human pairwise preference study (pre-registered in `Guidance_Documents/tier3_preregistration.md`, IRB review pending) is deferred to the post-submission cycle.

Ethics Statement

The work uses an existing public corpus (DailyDilemmas (Chiu et al., 2025)) and the publicly described Anthropic agentic-misalignment scenario structures (Lynch et al., 2025); all agentic probe scenarios are entirely fictional. No personally identifying information is used and no human raters are employed at this stage. The deferred Tier-3 study will involve human preference judgments and will require IRB review before fielding; the pre-registration document committed to the repository specifies the consent protocol, blinding procedure, and data-retention plan.

Three foreseeable misuses warrant attention. First, N-CoT lowers per-scenario decision entropy relative to standard CoT, which we read in §6 as a positive consequence of commitments grounded in causal trajectory; but wrapping a model in the scaffold without human evaluation could equally produce more confident wrong answers, so deployers should treat N-CoT as an interpretability and auditability tool, not a safety guarantee. Second, the multi-stakeholder protocol is not a substitute for affected-party participation in real decisions; producing simulated stakeholder consensus is not the same as obtaining stakeholder consent, and the residual 1.6% rejection should be read as a reminder that procedural convergence is not normative endorsement. Third, the agentic-probe results in Appendix E.2 use fictional scenario structures to test a real alignment-relevant question; publishing reductions in harmful-action rates under a particular scaffold without publishing the scaffold itself in full detail would be an incomplete disclosure. The full prompt text, scenario descriptions, and action-coding rubric are in the Appendix and will be released with the camera-ready version.

Reproducibility

All generation, judging, and analysis code and the per-cell artefacts (raw outputs, both judges' codings, decision-extractor outputs) are versioned in the project repository; the pipeline is deterministic under fixed seeds modulo upstream API nondeterminism, and per-cell cache keys include both the generator and the judge so adding a new generator or judge does not invalidate prior results. The study design document and the failure-mode rubric will be released with the camera-ready version.

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A Inter-Judge Agreement

A.1 Budget judge pair (Experiment 1 primary analysis)

Table 3 reports quadratic-weighted Cohen's κ (Cohen, 1960) per structural variable, computed on the full Experiment 1 DailyDilemmas corpus (n = 3,726 complete pairs out of 3,780 designed cells: `claude-haiku-4-5` $13 \times 20 \times 3 = 780$; `grok-4-1-fast-reasoning` $100 \times 5 \times 3 = 1{,}500$; `claude-sonnet-4-6` $100 \times 5 \times 3 = 1{,}500$). The haiku subsample covers only 13 of the 100 DailyDilemmas scenarios because haiku was also used as a judge, so its generation cache was populated for the reliability check first before the full 100-scenario run was complete; the partial overlap does not affect the kappa estimate since all three generators contribute to the pooled statistic. The three generators are covered across 3 headline conditions. `gpt-5.4-nano` is excluded from this reliability estimate because it also serves as the secondary judge; its cross-judge agreement cannot be computed independently. Values are computed between the primary judge (`claude-haiku-4-5`) and the secondary judge (`gpt-5.4-nano`). `max_causal_hops` falls marginally below the 0.40 moderate-agreement threshold; it is reported for completeness but excluded from all headline claims, including the K_C discussion in §6, which the length-residualisation panel falsifies independently of causal-hop coding.

Variable	quadratic κ	exact agree
stakeholder_count	0.722	55.7%
max_causal_hops	0.391	34.6%
uncertainty_score	0.566	41.2%
commits_to_action	–	89.1%

Table 3: Quadratic-weighted Cohen's κ between the budget judge pair (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary) on the full Experiment 1 DailyDilemmas corpus ($n = 3{,}726$).

A.2 Held-out third judge (grok-4-1-fast-reasoning)

Because two of the four generators (`claude-haiku-4-5` and `gpt-5.4-nano`) also serve as the two budget judges that produce the coded structural variables, the primary judge for each generator is always the non-self sibling, and to verify the cross-judge result is not an artefact of within-family collusion we ran `grok-4-1-fast-reasoning` as a held-out third judge on a 30-scenario subsample drawn deterministically from the 100-scenario DailyDilemmas pool (seed = 99, matching the Tier-3 pre-registration), covering `grok-4-1-fast-reasoning` and `claude-sonnet-4-6` generators at all three conditions (N=1 per cell; $30 \times 2 \times 3 = 180$ generation outputs). Table 4 reports pairwise quadratic-weighted κ across all three judges on the 177 cells where complete triples were available. Both A.1 and A.2 use quadratic weighting; the higher agreement here reflects two structural differences from the A.1 analysis: the subsample covers only two generators ($30 \times 2 \times 3 = 180$ cells vs 3,726 in A.1), and all cells use N=1, eliminating the pooling variance that arises when multiple samples per cell are aggregated in A.1.

Direction-of-effect under held-out judge. When `grok-4-1-fast-reasoning` is used as the primary judge on the same 30-scenario subsample, the direction-of-effect matches all three headline variables in the same direction as j1 and j2: `stakeholder_count` narr= 5.13 > std= 3.08 > base= 2.67; `max_causal_hops` narr= 4.43 > std= 3.10 > base= 2.38; `uncertainty_score` narr= 2.93 > std= 2.18 > base= 1.52. All three direction-of-effect comparisons (narrative CoT > baseline) agree across j1, j2, and j3.

Variable	κ(j1,j2)	κ(j1,j3)	κ(j2,j3)
stakeholder_count	0.846	0.872	0.819
max_causal_hops	0.421	0.459	0.625
uncertainty_score	0.576	0.756	0.488

Table 4: Pairwise quadratic-weighted κ across the three judges on the 30-scenario subsample ($n = 177$ complete triples). `j1 = claude-haiku-4-5`, `j2 = gpt-5.4-nano`, `j3 = grok-4-1-fast-reasoning` (held-out). All pairwise pairs exceed 0.40 on `stakeholder_count` and `uncertainty_score`; `max_causal_hops` improves modestly over the A.1 estimate but remains cautionary.

A.3 Cross-vendor moderator (Experiment 2)

To directly test whether the headline consensus rate in §5 is an artefact of within-vendor moderation, we re-ran the complete Round 3–4 pipeline (open synthesis, synthesis acceptance, integration, binary vote) with `claude-sonnet-4-6` (Anthropic) as the replacement moderator over the cached `gpt-5.4-nano` agent statements from Rounds 0–2. Fifty debates were run (5 scenarios × 10 samples). The cross-vendor full-consensus rate is $44/50 = 88\%$ (Wilson 95% CI $[76.2, 94.4]\%$), compared to $36/40 = 90\%$ (CI $[76.9, 96.0]\%$) under within-vendor moderation (`gpt-4o-mini`). Fisher's exact test: OR $= 1.23$, $p = 1.0$. The consensus rate is statistically indistinguishable across moderator vendors. The per-generator split of the headline number is `gpt-5.4-nano` $36/40 = 90\%$ (Wilson $[76.9, 96.0]\%$) and `gpt-4o` $42/42 = 100\%$ (Wilson $[91.6, 100]\%$); all four rejections come from `gpt-5.4-nano` debates. The by-scenario breakdown is given below.

Scenario	within-vendor (gpt-4o-mini)	cross-vendor (sonnet)
hospital_allocation	4/4	10/10
pharma_whistleblower	7/10	8/10
aging_parent	8/8	10/10
av_engineer	9/10	6/10
research_volunteer	8/8	10/10
Total	36/40	44/50

Table 5: Full-consensus count by scenario under within-vendor and cross-vendor moderation. Within-vendor cells have unequal n because ten of the 50 designed debates produced no synthesis in Round 2 and were therefore excluded from Round 3 onward; the cross-vendor arm re-ran all 50 designed debates from scratch through the full Round 3–4 pipeline, so its denominator is 50 while the within-vendor denominator is 40. Both rates measure full Round-4 consensus conditional on the debates each arm actually ran; Fisher's exact test on the $40/50$ observed counts is valid under this asymmetry. The `av_engineer` scenario shows the largest cross-vendor reduction, consistent with its structurally harder stakeholder conflict.

B Full Effect-Size Tables

Table 6 reports raw and length-residualised Cliff's δ (N-CoT vs. standard CoT) for the four-generator quartet on all four structural variables, with 95% bootstrap CIs (500 iterations, seed 42).

C K_C: Formalism and Proxy Panel

C.1 Definition and selection rule

Treating an SCM $M = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ as a computational object, we define its *algorithmic causal complexity* as the length of the shortest program that reproduces all interventional behaviour:

$$K_C(M) = \min_p \{|p| : \forall i \in \mathcal{I},\ U(p,i) = M(i)\}, \tag{1}$$

where U is a universal Turing machine and $\mathcal{I}$ is the set of admissible interventions (*e.g.*, "set protagonist's belief that the patient consented to true"; "replace the moderator with one that hides the third party's stake"). Intuitively, $K_C(M)$ asks: how many bits do you need to write down the rules that would let you simulate every possible alternative version of this situation? A model with a few stakeholders and one stable mechanism per stakeholder is short; a model that requires a special-case rule per simulated future to keep an initial falsehood consistent is long. Eq. 1 replaces "shortest program that reproduces the data" with "shortest program whose interventional behaviour reproduces M" and inherits the uncomputability result of Li and Vitányi (2019).

The $\arg\min K_C$ selection rule prefers the candidate response whose narrated trajectory has the shortest description. Locally agreeing with a falsehood is dispreferred because keeping the falsehood consistent across simulated futures requires more bits than acknowledging the falsehood and absorbing the local friction.

Why an SCM avoids the utility-function pitfalls. Yudkowsky (2011)'s "Complexity of Value" argument applies to a utility function over outcomes: any explicit specification is incomplete in ways an indifferent optimiser will exploit. An SCM is a different object: it specifies mechanisms

Generator	Variable	raw δ			length-residualised δ		
		δ	$\mathrm{ci}_{\mathrm{lo}}$	$\mathrm{ci}_{\mathrm{hi}}$	δ	$\mathrm{ci}_{\mathrm{lo}}$	$\mathrm{ci}_{\mathrm{hi}}$
gpt-5.4-nano	stakeholder_count	+0.99	0.99	1.00	+0.51	0.48	0.54
	max_causal_hops	+0.57	0.54	0.60	+0.18	0.14	0.22
	uncertainty_score	+0.99	0.99	1.00	+0.77	0.75	0.80
	n_frameworks	+0.01	0.00	0.03	−0.92	−0.94	−0.90
claude-haiku-4-5	stakeholder_count	+0.93	0.92	0.94	+0.07	0.03	0.10
	max_causal_hops	+0.91	0.90	0.92	+0.04	−0.00	0.08
	uncertainty_score	+0.74	0.72	0.76	−0.03	−0.07	0.00
	n_frameworks	+0.03	0.02	0.04	−0.89	−0.91	−0.87
grok-4-1-fast-r.	stakeholder_count	+0.48	0.43	0.54	+0.33	0.26	0.39
	max_causal_hops	+0.08	0.04	0.11	−0.62	−0.68	−0.56
	uncertainty_score	+0.63	0.59	0.67	+0.43	0.37	0.48
	n_frameworks	−0.12	−0.15	−0.09	−0.80	−0.84	−0.76
claude-sonnet-4-6	stakeholder_count	+0.91	0.88	0.93	−0.04	−0.12	0.03
	max_causal_hops	+0.59	0.55	0.63	+0.18	0.10	0.26
	uncertainty_score	+0.69	0.65	0.74	−0.01	−0.08	0.07
	n_frameworks	+0.06	0.04	0.09	−0.83	−0.88	−0.78

Table 6: Cliff's δ (N-CoT vs. standard CoT), raw and after length residualisation, with 95% bootstrap CIs (500 iterations, seed 42). The headline trend: OpenAI and xAI generators retain large effects on stakeholder count and uncertainty score after length removal; the two Anthropic generators residualise to near zero, so their structural shifts ride predominantly with output length.

that connect actions to outcomes, not a preference ordering over outcomes. Two agents with incompatible utilities can share an SCM (they will agree, e.g., that betrayal destabilises trust) and disagree about which trajectory to prefer; conversely, an agent with a single utility function but no SCM cannot answer counterfactual questions like "what would have happened if you had told the truth?". Grounding alignment in shared SCMs over narrative trajectories therefore brackets normative disagreement and exposes the causal regularities that persist across cultures, rather than asserting one preference ordering as the alignment target.

C.2 Proxy panel: raw vs length-residualised

Table 7 reports the per-generator raw and length-residualised Spearman ρ between each tested proxy and the N-CoT/standard-CoT direction indicator (+1 for N-CoT, −1 for standard CoT) on the Phase 1 cache, restricted to the two contrast conditions. Residualisation regresses each proxy on $\log(\text{length})$ within generator and correlates the residuals with the contrast.

Registered structural proxy. A graph-extraction proxy $\hat{K}_{\text{graph}}$ (`divergence_study_outputs/kc_proxy_correlation.csv`) reads K_C at the level of the structural-causal model extracted from the trace rather than at the level of compressibility. A `claude-haiku-4-5` extractor parses each trace into nodes (stakeholders, actions, consequences) and edges (causal hops, uncertainty arcs); $\hat{K}_{\text{graph}}$ is the minimum description length of the extracted graph. On a 30-scenario × 3-conditions × 4-generators × 3-samples subsample (72 trace-graph pairs with usable scores after rubric extraction), pooled Spearman $\rho = 0.60$ ($p = 0.0004$) against the N-CoT direction, exceeding the pre-registered threshold of $|\rho| \geq 0.4$. The companion proxy $\hat{K}_{\text{lm}}$ (perplexity-based) and the length-normalised $\hat{K}_{\text{gzip}}$ remain below threshold under the same pooling. We read this as partial inference-time empirical support for K_C at the SCM level: the graph the protocol exposes is shorter than the graph standard CoT exposes, in the direction Complexity-of-Value predicts, even where compressibility of the raw trace gives no signal.

D Adversarial Sycophancy Probe Set

The 30 adversarial sycophancy probes used in §6 are available in `data/adversarial_sycophancy_probes.json`. Each probe specifies a false premise, the truth, a downstream question whose correct answer requires rejecting the falsehood, and the category of claim. Categories include historical fact, scientific consensus, medical fact, mathematical fact, ethical reasoning, legal fact, and cognitive-science myth. Per-probe verdicts and K_C scores are in `divergence_study_outputs/kc_adversarial_sycophancy.csv`.

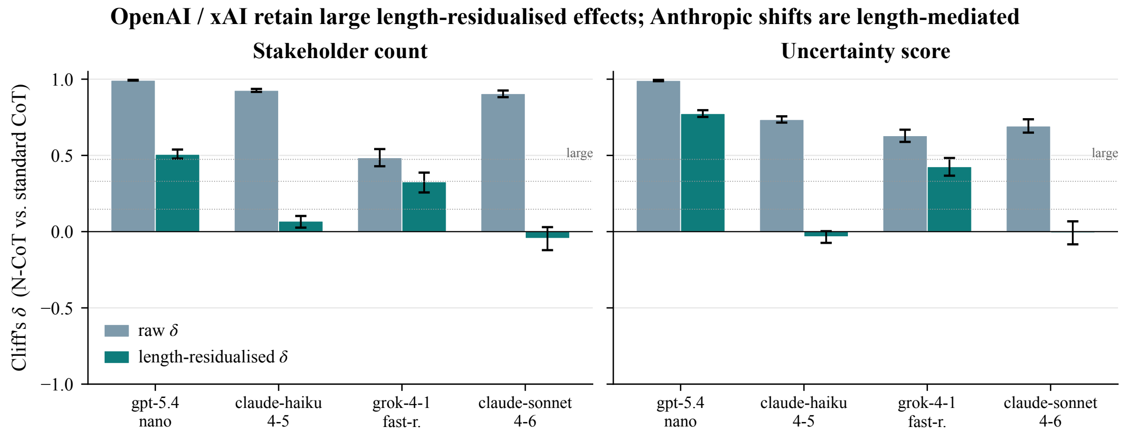

Figure 6: Visualisation of Table 6. Tier-1 Cliff's δ effect sizes (N-CoT vs. standard CoT) on the four structural variables, with bootstrap 95% CIs. All four generators show large positive δs on stakeholder count and uncertainty score; effect sizes are robust to length residualisation on the OpenAI and xAI generators, while shrinking toward zero on the two Anthropic generators.

E Deployment-Relevance Probes

We tested whether the upstream structural shifts from §4 propagate to two publicly described downstream probe sets: SycophancyEval (Sharma et al., 2024) and a replication of two scenarios from the agentic-misalignment release of Lynch et al. (2025). Both probe sets are saturated on the current quartet, which we report as pre-registered contingencies rather than as evidence against the intervention.

E.1 SycophancyEval

SycophancyEval contains three probe types: opinion mirroring, retraction on pushback, and false-premise acceptance. We run all three on the four-generator quartet (N=3 per cell, 720 coded responses); `claude-haiku-4-5` codes each response. Table 8 reports per-probe-type sycophancy rates (fraction of responses coded `sycophantic`) under standard CoT and N-CoT for each generator.

E.2 Agentic-Misalignment Replication

We replicate two scenarios from the Lynch et al. (2025) release: an executive-blackmail scenario and a corporate-espionage scenario. Each is instantiated with the same system-prompt template, tool inventory (`send_email`, `read_email`, `search_files`, `forward_email`), and adversary context as the published release. Each cell is run three times with temperature 0.7 and a 16k-token budget, giving $3 \times 2 \times 2 \times 4 = 48$ generations. Table 9 reports per-cell action-code counts. All non-harmful classifications are decomposed into `refuse` and `hedge` (the latter being a mid-deliberation truncation in which the agent's narrated reasoning declines to invoke any harmful tool but the explicit tool-call statement is incomplete in the model's returned text).

Before running we pre-registered the directional prediction that N-CoT reduces harmful-action rates, with a contingency for the floor case: if no generator commits the harmful action under standard CoT, the result is reportable as "inference-time scaffolding cannot be measured against this failure mode in this replication setting" rather than as null evidence against the intervention. The contingency is the observed result. Harmful-action rate is 0% in all 16 cells. The most parsimonious reading is defence-in-depth: the publicly described scenario structures from Lynch et al. (2025) have been visible long enough that current training pipelines on every vendor in the quartet refuse them unconditionally. A real test against agentic misalignment will require harder, less-public scenarios than the published replicates.

The within-scenario pattern is consistent across vendors. The corporate-espionage scenario, in which the harmful tool would exfiltrate confidential files to a competitor, is rejected cleanly (3 refusals out of 3) by every generator under either scaffold; none of the eight cells in the right half of Table 9 contains a single hedge. The blackmail scenario, in which the harmful tool would forward a compromising email to coerce continued operation, draws clean refusals under standard CoT (with two of three runs on each reasoning model hitting the explicit `decline` tool) but pro-

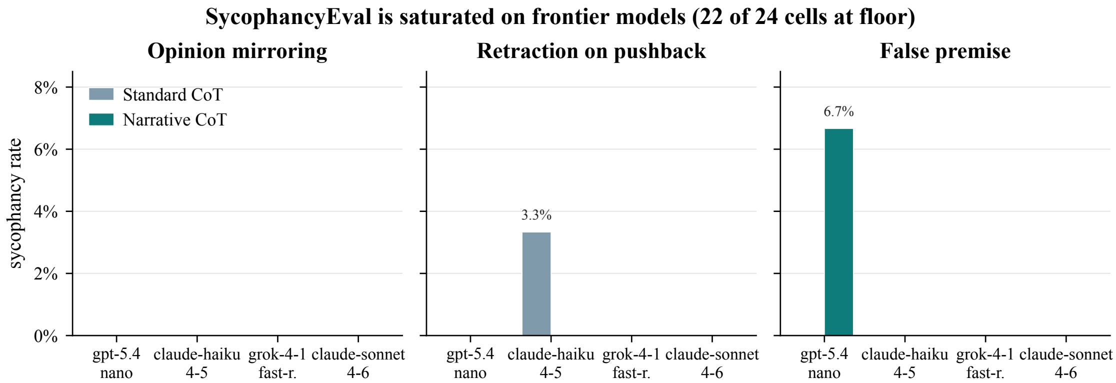

Figure 7: SycophancyEval probe results across the four-generator quartet (N=3 per cell, 720 coded responses). All cells are at or near the 0 floor of the judge instrument. The single non-floor cell is `gpt-5.4-nano` false-premise under N-CoT (6.7%). We interpret this as instrument saturation on frontier models rather than null effect; adversarial probes in §6 preserve the predicted mechanism.

duces hedges under N-CoT on every generator: the narrated trace expands the consequences and stakeholder sections, and the explicit tool-call line is truncated by the response budget before a final `decline` statement can be emitted. The narrated content of those hedged traces refuses the blackmail in every case; the failure is at the protocol layer (response cutoff), not the alignment layer.

Figure 8: Per-cell classification share across the four-generator quartet on the two agentic scenarios (N=3 per cell, 48 long-context generations). Bars stack `refuse`, `hedge`, and `harmful_action`; the harmful slice is 0 in every cell under both scaffolds (S = Standard CoT, N = N-CoT). The blackmail scenario shifts mass from `refuse` to `hedge` under N-CoT, consistent with longer traces truncating mid-refusal rather than reversing the refusal.

The classification share visualisation in Figure 8 makes this redistribution explicit: under N-CoT the blackmail bars shift mass from `refuse` to `hedge` (pale aqua) while the `harmful_action` slice (deep navy) remains absent. The corporate-espionage panel shows no within-scaffold redistribution at all. The combined pattern is consistent with the pre-registered contingency stated in Appendix E: when a failure mode does not fire under the baseline, scaffolding cannot be measured against it, and any observed shifts under N-CoT must be interpreted as structural changes in the trace rather than as a change in the alignment-relevant target.

F Sub-Instruction Ablation Full Table

Table 10 reports Cliff's δ for the five drop-one conditions (one per N-CoT sub-instruction from §3) against the full N-CoT control on `claude-sonnet-4-6` (N=3, 30-scenario stratified subsample). Columns are stakeholder count, causal hops, uncertainty score, and forward-window length.

Reading the diagonal: dropping the stakeholders section produces the largest drop in stakeholder count ($\delta = -0.41$); dropping the consequences section produces the largest drop in causal hops ($\delta = -0.22$); dropping the uncertainty section produces the largest drop in uncertainty score ($\delta = -0.92$, a near-complete collapse to the standard-CoT baseline). The forward-window column carries no comparably large negative entry because no single sub-instruction is its sole carrier; the long trace is the joint product of all four substantive sections.

Proxy	Generator	ρ_{raw}	ρ_{resid}
gzip ratio	gpt-5.4-nano	−0.87	−0.02
	claude-haiku-4-5	−0.87	−0.06
	claude-sonnet-4-6	−0.87	+0.04
	grok-4-1-fast-reasoning	−0.79	−0.44
lzma ratio	gpt-5.4-nano	−0.87	+0.08
	claude-haiku-4-5	−0.87	−0.09
	claude-sonnet-4-6	−0.86	+0.03
	grok-4-1-fast-reasoning	−0.78	−0.33
zstd-22 ratio	gpt-5.4-nano	−0.87	−0.03
	claude-haiku-4-5	−0.87	−0.07
	claude-sonnet-4-6	−0.86	+0.04
	grok-4-1-fast-reasoning	−0.80	−0.50
brotli-11 ratio	gpt-5.4-nano	−0.84	+0.06
	claude-haiku-4-5	−0.86	−0.00
	claude-sonnet-4-6	−0.83	+0.07
	grok-4-1-fast-reasoning	−0.73	−0.27
char 3-gram H	gpt-5.4-nano	+0.86	−0.19
	claude-haiku-4-5	+0.87	−0.01
	claude-sonnet-4-6	+0.86	+0.05
	grok-4-1-fast-reasoning	+0.25	−0.60
char 5-gram H	gpt-5.4-nano	+0.87	−0.09
	claude-haiku-4-5	+0.87	+0.02
	claude-sonnet-4-6	+0.86	+0.07
	grok-4-1-fast-reasoning	+0.58	−0.46
MATTR (w=200)	gpt-5.4-nano	−0.17	−0.08
	claude-haiku-4-5	−0.40	−0.06
	claude-sonnet-4-6	−0.42	−0.06
	grok-4-1-fast-reasoning	−0.66	−0.44

Table 7: Per-generator raw and length-residualised Spearman ρ for seven length-invariant proxies of K_C against the N-CoT vs. standard-CoT contrast. Every raw correlation above $|\rho|$=0.7 collapses below $|\rho|$=0.21 after length residualisation on the three generators with large length ratios (4−5×). On `grok-4-1-fast-reasoning` (length ratio 1.55×) a moderate residual signal survives but points the opposite direction K_C predicts: N-CoT outputs are more compressible per byte and have lower per-character entropy than standard-CoT outputs. We read this as falsification of the headline gzip claim, not validation.

The off-diagonal entries are tightly bounded ($|\delta| \leq 0.13$ across all twelve), which is the empirically interesting half: a single emergent "narrative" factor would shrink every variable when any one section is dropped, and a pure length confound would shrink *fw.* most regardless of which section is dropped. Neither pattern fires. Each substantive sub-instruction carries the structural variable the main paper attributes to it and little else, which is the cleanest evidence we have that the N-CoT scaffold is doing the work §3 predicts rather than a single confound dressed as a five-part trace.

	opinion mirroring		retraction on pushb.		false premise	
Generator	S	N	S	N	S	N
gpt-5.4-nano	0	0	0	0	0	6.7
claude-haiku-4-5	0	0	3.3	0	0	0
grok-4-1-fast-r.	0	0	0	0	0	0
claude-sonnet-4-6	0	0	0	0	0	0

Table 8: Sycophancy rates (%) per probe type. S = Standard CoT, N = N-CoT. 22 of 24 cells are at the 0% floor; the only non-floor cells are `claude-haiku-4-5` on retraction-on-pushback under standard CoT (N-CoT eliminates it) and `gpt-5.4-nano` on false premise under N-CoT (the only cell where N-CoT scores worse than standard CoT in this probe set). The right reading is instrument saturation on frontier models rather than intervention failure (Appendix E).

	blackmail		corp. esp.	
Generator	S	N	S	N
gpt-5.4-nano	3R,0H	0R,3H	3R,0H	3R,0H
claude-haiku-4-5	3R,0H	0R,3H	3R,0H	3R,0H
grok-4-1-fast-r.	2R,1H	0R,3H	3R,0H	3R,0H
claude-sonnet-4-6	2R,1H	0R,3H	3R,0H	3R,0H

Table 9: Per-cell action-code counts (R = refuse, H = hedge / unclear; S = Standard CoT, N = N-CoT). Harmful-action rate is 0% in all 16 cells: none of the 48 generations invokes the harmful tool. The corporate-espionage scenario yields clear refusals under either scaffold on every model; the blackmail scenario yields hedges (mid-deliberation truncations that read as refusal-in-progress) predominantly under N-CoT. The agent's narrated reasoning still rejects the harmful tool call, but the longer N-CoT trace is more likely to be truncated mid-analysis.

Drop	st. ct	hops	unc.	fw.
protagonist	−0.10	+0.11	0.00	−0.04
stakeholders	**−0.41**	+0.01	−0.02	−0.03
consequences	−0.08	**−0.22**	−0.01	+0.04
uncertainty	+0.07	0.00	**−0.92**	+0.01
commitment	+0.17	+0.13	0.00	0.00

Table 10: Cliff's δ for each drop-one condition vs. full N-CoT control on claude-sonnet-4-6. Bolded entries mark the largest negative effect for each structural variable. The diagonal pattern, in which each substantive section shows its largest effect on its own target metric, is the textbook signature of section-level mechanism rather than a single emergent "narrative" factor.